

DEVELOPMENT OF THE ATOMIC MODEL

KEY VOCABULARY

Plum pudding: Thomson's model: a ball of positive charge with electrons in it.

Nuclear model: a tiny dense positive nucleus surrounded by electrons.

Alpha scattering: experiment that proved most of the atom is empty space.

Energy levels: Bohr's shells where electrons orbit at set distances.

COMMON MISCONCEPTIONS

~~— The atom was discovered in one experiment.~~

✓ The model changed over time as new evidence appeared.

~~— Electrons were known before the nucleus.~~

✓ Correct — but the nucleus was found later by Rutherford.

~~— Neutrons were in the first models.~~

✓ Chadwick discovered the neutron about 20 years later.

HOW THE MODEL CHANGED



Dalton
solid sphere



Thomson
plum pudding



Rutherford
nuclear model



Bohr
electron shells

time / new evidence →

1 PLUM PUDDING

Describe the **plum pudding model** of the atom. [2 marks]

2 ALPHA SCATTERING

Most alpha particles passed straight through the gold foil. **Explain** what this showed. [2 marks]

3 DEFLECTED PARTICLES

A few alpha particles bounced straight back. **Explain** what this told scientists. [2 marks]

4 BOHR'S IDEA

State how Bohr changed the nuclear model. [1 mark]

5 ORDER THE MODELS

Complete the order using the word bank. [2 marks]

solid sphere → ____ → *nuclear model* → ____ (*shells*).

Dalton

Thomson

Rutherford

Bohr

6 WHY MODELS CHANGE

Explain why scientific models are changed over time. [2 marks]
